

# Extremal Seidel Quadratic Forms on the Boolean Cube

Bounds, Exact Values, Barriers, and Open Problems

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3 August 2026

## Abstract

For  $n \geq 2$ , define

$$F(n) = \min_{a_{ij} \in \{-1,1\}} \max_{x \in \{-1,1\}^n} \left| \sum_{1 \leq i < j \leq n} a_{ij} x_i x_j \right|, \quad L_n = F(n) n^{-3/2}.$$

The existence and value of  $\lim_n L_n$  remain open. This report records a set of internally audited bounds, structural identities, exact computations, counterexamples, and continuation problems. The main universal result is

$$0.336493364432 \dots \leq \liminf_{n \rightarrow \infty} L_n \leq \limsup_{n \rightarrow \infty} L_n \leq \frac{1}{2}.$$

The lower endpoint follows from a field-plus-spin rounding, independently reconstructed here after its appearance in an external AI-assisted research ledger; the upper endpoint follows from Paley conference matrices and principal restriction. The earlier squared-Gram argument remains as an explicit finite-order bound. Exact values are reported through order 15. A second part of the report maps the remaining absolute-field problem: it is proved in several diffuse, bounded-component, bounded-degree, and twin-fibre regimes, while a genuinely dense overlapping-correlation regime remains. The report also records barriers that rule out several natural spectral, compositional, superadditive, and low-degree Fourier approaches. For composition, it is enough to prove a power-saving defect on comparable block sizes; explicit rectangular estimates already give sublinear defects when one block is  $o(n)$  and power savings when it is polynomially smaller. A quotient-stratified shell theorem interpolates exactly between marginal histograms and exhaustive bridge response, with a multiplicity bound showing when such compression is impossible. Its global two-replica test is an addition-fiber collision inequality with sharp occupancy bounds, exposing both a symmetry reduction and a Fourier-dilution obstruction.

**Status.** This is an independent research report, not a journal-refereed paper. Statements are labelled as proved, exhaustive computation, numerical, open, or refuted. “Internally audited” means that normalizations and stated dependencies were re-derived or checked independently during the project; it does not replace external peer review. The public repository reproduces its own exact search only through order 6. A pinned external package for order 11 was rerun locally, as were external witnesses and solver cases through order 14; the remaining larger results retain their historical evidence labels. External source code is cited rather than vendored.

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## 1 Problem, normalization, and interpretations

**Definition 1.1.** A *Seidel matrix* of order  $n$  is a matrix  $A \in \mathbb{R}^{n \times n}$  such that  $A = A^\top$ ,  $A_{ii} = 0$ , and  $A_{ij} \in \{-1, 1\}$  for  $i \neq j$ . The set of such matrices is denoted by  $\mathcal{S}_n$ .

For  $A \in \mathcal{S}_n$  and  $x \in \{-1, 1\}^n$ , put

$$Q_A(x) = \sum_{i < j} A_{ij} x_i x_j = \frac{1}{2} x^\top A x, \quad M(A) = \max_x |x^\top A x|.$$

Also write

$$M_+(A) = \max_x x^\top A x, \quad M_-(A) = -\min_x x^\top A x.$$

The normalization ledger is therefore

$$F(n) = \frac{1}{2} \min_{A \in \mathcal{S}_n} M(A), \quad L_n = \frac{F(n)}{n^{3/2}}. \quad (1)$$

Every factor of two in this report is with respect to (1).

The problem has several equivalent or neighbouring interpretations. It is a discrepancy problem for a two-colouring of  $E(K_n)$ , a deterministic Ising ground-state minimization, and the minimum  $L^\infty$  norm of a homogeneous order-two Rademacher chaos. The asymptotic question

was posed by Ivanisvili on MathOverflow [10]. Volberg’s Boolean Sidon quantity  $T_{n,2}$  is exactly  $F(n)$  [15]; Astashkin and Lykov establish the  $n^{3/2}$  order up to universal constants in a more general weighted setting [1]. The induced-subset discrepancy studied by Erdős and Spencer is related but not identical [5].

Switching  $A \mapsto DAD$ , where  $D$  is diagonal with entries  $\pm 1$ , and simultaneous row/column permutation both preserve  $M$ ,  $M_{\pm}$ , and the cut quantity below. Thus the natural finite objects are switching classes (two-graphs), modulo permutation when classification is desired.

## 2 Principal conclusions

Let  $\phi$  and  $\Phi$  be the standard normal density and distribution function. Let  $t_* > 0$  maximize  $4\phi(t)(2\Phi(t) - 1)$ , equivalently

$$2\phi(t_*) = t_*(2\Phi(t_*) - 1), \quad t_* = 0.8769009855\dots,$$

and put

$$\gamma_* := 2\phi(t_*)(2\Phi(t_*) - 1) = 0.336493364432\dots$$

**Theorem 2.1** (universal window). *For every  $n \geq 2$  and every  $A \in \mathcal{S}_n$ ,*

$$M(A) \geq \frac{2}{\pi}n(n-1) \arcsin\left(\frac{1}{\sqrt{n-1}}\right) \geq \frac{2}{\pi}n\sqrt{n-1}. \quad (2)$$

*In addition, every sequence  $A_n \in \mathcal{S}_n$  satisfies*

$$\liminf_{n \rightarrow \infty} \frac{M(A_n)}{n^{3/2}} \geq 2\gamma_*.$$

*There are Seidel matrices of every sufficiently large order, after principal restriction from nearby Paley conference orders, for which*

$$F(n) \leq \left(\frac{1}{2} + o(1)\right) n^{3/2}.$$

*Hence*

$$\gamma_* \leq \liminf L_n \leq \limsup L_n \leq \frac{1}{2}.$$

The existence of the limit is not asserted. The exact data do not presently distinguish convergence from persistent arithmetic oscillation.

### 2.1 Exact values

The following values are classified as **EXHAUSTIVE COMPUTATION**. Orders 2 through 6 are regenerated by the public code. Order 11 now has a separately published standalone enumeration that was rerun during this audit; public external witness and solver reruns also corroborate orders 12 through 14. The remaining entries above order 6 retain the historical exhaustive or threshold-emptiness evidence described in Section 6.

| $n$           | 2 | 3 | 4 | 5 | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
|---------------|---|---|---|---|----|----|----|----|----|----|----|----|----|----|
| $\min_A M(A)$ | 2 | 6 | 8 | 8 | 10 | 18 | 20 | 24 | 26 | 34 | 36 | 40 | 42 | 54 |
| $F(n)$        | 1 | 3 | 4 | 4 | 5  | 9  | 10 | 12 | 13 | 17 | 18 | 20 | 21 | 27 |

The minimizer is unique up to switching and permutation at orders 12, 13, and 14: respectively a doubling of  $C_6$ , a vertex deletion of  $C_{14}$ , and the Paley conference matrix  $C_{14}$ . At order 15, the all-positive one-vertex extension of  $C_{14}$  is optimal with  $M = 54$ . Monotonicity and congruence imply  $\min_A M(A) \geq 56$  at order 16.

For the cut discrepancy introduced below, the exact minimum for  $n = 3, \dots, 14$  is

$$2, 4, 4, 5, 8, 10, 12, 13, 16, 18, 20, 21.$$

At order 15 it is either 24 or 26; the exhaustive route needed to separate the two exceeded the available memory.

### 3 Core identities and lower bounds

**Lemma 3.1** (fill-in). *For  $A \in \mathcal{S}_n$  and  $u \in \{0, \pm 1\}^n$ ,  $|u^\top A u| \leq M(A)$ .*

*Proof.* Complete the zero coordinates of  $u$  by independent Rademacher signs to obtain  $x \in \{\pm 1\}^n$ . All mixed and off-support terms have expectation zero; no diagonal term survives because  $A_{ii} = 0$ . Thus  $u^\top A u = \mathbb{E}[x^\top A x]$ , whose absolute value is at most  $M(A)$ .  $\square$

The zero diagonal is essential. For  $\begin{pmatrix} 4 & -1 \\ -1 & -3 \end{pmatrix}$ , the Boolean maximum is 3, but the vector  $(1, 0)$  has quadratic value 4.

For a partition  $[n] = S \sqcup T$ , define

$$\text{Cut}(A) = \max_{S \sqcup T = [n]} \max_{u \in \{\pm 1\}^S, v \in \{\pm 1\}^T} u^\top A_{S,T} v.$$

**Theorem 3.2** (cut identity). *For every  $A \in \mathcal{S}_n$ ,*

$$M_+(A) + M_-(A) = 4 \text{Cut}(A).$$

*In particular,  $\max_x |Q_A(x)| \geq \text{Cut}(A)$ .*

*Proof.* For independently chosen  $x, y \in \{\pm 1\}^n$ , put  $u = (x + y)/2$  and  $w = (x - y)/2$ . Their supports partition  $[n]$  and

$$x^\top A x - y^\top A y = 4u^\top A w.$$

Conversely every signed cut arises from such a pair. Maximizing  $x$  and  $y$  independently gives the identity.  $\square$

**Proposition 3.3** (fixed-split first moment). *For every  $1 \leq s < n$ ,*

$$\text{Cut}(A) \geq (n - s) \mathbb{E}|S_s|,$$

*where  $S_s$  is a sum of  $s$  independent Rademacher variables.*

*Proof.* Fix  $|S| = s$ . For  $u \in \{\pm 1\}^S$ , optimize the signs on  $T$  to obtain  $\|A_{S,T}^\top u\|_1$ , then average over  $u$ . Each of the  $n - s$  coordinates is distributed as  $S_s$ .  $\square$

For odd  $s = 2k + 1$ ,

$$\mathbb{E}|S_s| = (2k + 1)4^{-k} \binom{2k}{k} \geq \sqrt{\frac{2}{\pi}} \sqrt{s}.$$

The parity restriction matters: the displayed inequality is false for even  $s$ . A sharpened Wallis inequality due to Kazarinoff [11], together with a six-residue finite check, absorbs the rounding loss in choosing an odd integer near  $n/3$ . This gives the finite theorem

$$F(n) \geq \max_{1 \leq s < n} (n - s) \mathbb{E}|S_s| \geq \frac{2\sqrt{2}}{3\sqrt{3}\pi} n^{3/2} = 0.3071059106 \dots n^{3/2}. \quad (3)$$

The stronger asymptotic bound is Theorem 2.1.

### 3.1 Two-orientation squared-Gram rounding

Put  $q = n - 1$  and  $B = A/\sqrt{q}$ . Since  $B_{ii} = 0$  and  $(B^2)_{ii} = 1$ , the matrices

$$X_\sigma = \frac{1}{2}(I + \sigma B)^2, \quad \sigma \in \{-1, 1\},$$

are positive semidefinite correlation matrices. Hyperplane-round each  $X_\sigma$ . For  $i \neq j$ , write  $b = B_{ij}$  and  $c = (B^2)_{ij}/2$ . The sign-correlation identity yields terms involving  $\arcsin(c + \sigma b)$ . Averaging the two orientations uses

$$b[\arcsin(c + b) - \arcsin(c - b)] \geq 2|b| \arcsin |b|.$$

Indeed, the even density  $(1 - t^2)^{-1/2}$  has least mass on a fixed-length interval when that interval is centered at zero. Summing the  $n(n - 1)$  identical values  $|b| = q^{-1/2}$  proves (2).

### 3.2 Field-plus-spin improvement

The rounding scheme in this subsection appeared in the independent research ledger [2]. The proof below was reconstructed from the definitions, with the normalization, pair expansion, replacement error, and final factor of two checked afresh.

**Lemma 3.4** (spectral bootstrap). *For every  $A \in \mathcal{S}_n$ ,*

$$M(A) \geq \frac{\|A\|_2^3}{n}.$$

*Proof.* Let  $Av = \lambda v$ , where  $\|v\|_2 = 1$  and  $|\lambda| = \|A\|_2$ . The eigenvector equation gives

$$|\lambda|\|v\|_\infty \leq \|v\|_1 \leq \sqrt{n}.$$

Put  $c = \|v\|_\infty^{-1}$  and choose independent signs  $X_i$  with  $\mathbb{E}X_i = cv_i$ . The zero diagonal makes all variance terms disappear, so

$$\mathbb{E}X^\top AX = c^2 v^\top Av.$$

Taking absolute values and using  $c \geq \|A\|_2/\sqrt{n}$  proves the claim.  $\square$

*Proof of the field-plus-spin part of Theorem 2.1.* It is enough to consider a sequence for which  $M(A) = O(n^{3/2})$ ; otherwise the claimed lower bound is automatic along the relevant subsequence. Put  $m = n - 1$  and  $\lambda = \|A\|_2$ . Lemma 3.4 gives  $\lambda = O(n^{5/6})$ . For  $i \neq j$ , set

$$r_{ij} = \frac{A_{ij}(A^2)_{ij}}{m}.$$

Since  $\text{tr } A^2 = nm$ ,

$$\frac{1}{nm} \sum_{i \neq j} r_{ij}^2 = \frac{\|A^2 - mI\|_F^2}{nm^3} \leq \frac{\text{tr } A^4}{nm^3} \leq \frac{\lambda^2}{m^2} = O(n^{-1/3}). \quad (4)$$

Thus the normalized correlations  $r_{ij}$  tend to zero on average in absolute value.

Fix  $t > 0$  and  $\tau > 0$ . Let  $\xi_1, \dots, \xi_n$  be independent Rademacher signs, let  $Z_1, \dots, Z_n$  be independent standard Gaussians, and for  $\sigma \in \{-1, 1\}$  define

$$Y_i^\sigma = \text{sign} \left( \sigma \frac{(A\xi)_i}{\sqrt{m}} + t\xi_i + \tau Z_i \right). \quad (5)$$

Write

$$\psi_\tau(s) = 2\Phi(s/\tau) - 1,$$

and introduce the bounded smooth functions

$$\alpha(u) = \frac{\psi'_\tau(u+t) + \psi'_\tau(u-t)}{2}, \quad \beta(v) = \frac{\psi_\tau(v+t) - \psi_\tau(v-t)}{2}.$$

We record the pair calculation. Fix  $i \neq j$  and switch the matrix so that  $A_{ij} = 1$  and the common coefficients in row  $i$  are all 1. If the corresponding row- $j$  coefficients are  $c_k \in \{-1, 1\}$ , put

$$U = \frac{1}{\sqrt{m}} \sum_{k \neq i, j} \xi_k, \quad V = \frac{1}{\sqrt{m}} \sum_{k \neq i, j} c_k \xi_k, \quad d = \sum_{k \neq i, j} c_k.$$

Here  $d = A_{ij}(A^2)_{ij}$  before switching. After integrating out the Gaussian dithers in (5), Taylor expansion in  $\varepsilon = m^{-1/2}$  shows that the two orientations have equal zeroth order terms and

$$A_{ij} \left( \mathbb{E} Y_i^+ Y_j^+ - \mathbb{E} Y_i^- Y_j^- \right) = 4\varepsilon \mathbb{E}[\alpha(U)\beta(V)] + O_{t,\tau}(\varepsilon^2). \quad (6)$$

The factor 4 comes from the two endpoint spins and the two opposite orientations. All derivatives used in the remainder are bounded because  $\tau$  is fixed.

The pair  $(U, V)$  is a sum of  $m - 1$  independent two-dimensional Rademacher vectors, has coordinate variances  $1 - 1/m$ , and covariance  $d/m = r_{ij}$ . A two-dimensional Lindeberg replacement therefore has uniform error  $O_{t,\tau}(m^{-1/2})$ : each summand has size  $O(m^{-1/2})$ , there are  $O(m)$  summands, and third derivatives are bounded. Adjusting the variances to one costs  $O_{t,\tau}(m^{-1})$ . Gaussian covariance interpolation then gives, uniformly even near singular covariance,

$$\mathbb{E}[\alpha(U)\beta(V)] = \bar{\alpha} \bar{\beta} + O_{t,\tau}(m^{-1/2} + |r_{ij}|), \quad (7)$$

where, for  $G \sim N(0, 1)$  and  $\phi_v(s) = (2\pi v)^{-1/2} e^{-s^2/(2v)}$ ,

$$\bar{\alpha} = \mathbb{E}\alpha(G) = 2\phi_{1+\tau^2}(t), \quad \bar{\beta} = \mathbb{E}\beta(G) = 2\Phi\left(\frac{t}{\sqrt{1+\tau^2}}\right) - 1.$$

Average  $\sigma$  uniformly in (5). For every realization,  $\sigma(Y^\sigma)^\top A Y^\sigma \leq M(A)$ . Summing (6)–(7) over the  $nm$  ordered pairs, and using (4) to make the average of  $|r_{ij}|$  vanish, yields

$$\frac{M(A)}{n\sqrt{m}} \geq 4\phi_{1+\tau^2}(t) \left[ 2\Phi\left(\frac{t}{\sqrt{1+\tau^2}}\right) - 1 \right] - o(1).$$

First let  $n \rightarrow \infty$ , then let  $\tau \downarrow 0$ , and finally optimize over  $t > 0$ . This proves

$$\liminf_n \frac{M(A_n)}{n^{3/2}} \geq 4\phi(t_*)(2\Phi(t_*) - 1) = 2\gamma_*.$$

Choosing a minimizing  $A_n$  and applying (1) gives the asserted lower bound for  $L_n$ .  $\square$

### 3.3 Spectral-edge rounding

Let  $\mu(A) = \min\{\lambda_{\max}(A), |\lambda_{\min}(A)|\}$ . Applying hyperplane rounding to  $I + sA$  and to  $I - sA$  at their positive-semidefinite spectral endpoints gives

$$M(A) \geq \frac{2}{\pi} n(n-1) \arcsin\left(\frac{1}{\mu(A)}\right). \quad (8)$$

There is no sign cancellation because  $A_{ij} \arcsin(sA_{ij}) = \arcsin s$ . The bound is exact for  $A = J - I$ .

## 4 Upper bound and conference matrices

For  $q \equiv 1 \pmod{4}$  a prime power, the symmetric Paley conference matrix  $C$  of order  $q + 1$  satisfies

$$C^2 = qI, \quad \|C\|_2 = \sqrt{q}, \quad M(C) \leq (q + 1)\sqrt{q}.$$

Primes congruent to 1 modulo 4 have relative gaps  $o(x)$ ; the fixed modulus statement follows, for example, from Thorner–Zaman [14]. Restricting a nearby Paley matrix to a principal  $n \times n$  block and applying Lemma 3.1 proves the upper half of Theorem 2.1.

Conference matrices are spectrally flat, but their Boolean maxima are not uniformly close to the hyperplane-rounding value. An internally audited local-field calculation gives the following stronger statement.

**Theorem 4.1** (conference local improvement). *For every sequence of symmetric conference matrices,*

$$\liminf_n \frac{M(C_n)}{n\sqrt{n-1}} \geq \gamma_{\text{loc}} = 0.6894707435 \dots > \frac{2}{\pi}.$$

*If  $A_n$  differs from  $C_n$  in  $h_n = o(n^{3/2})$  unordered edges, the same coefficient holds for  $A_n$ .*

The stability assertion follows from the exact Lipschitz estimate

$$|M(A) - M(C)| \leq 4h.$$

For the local improvement, puncture the normalized conference matrix in the all-ones direction, hyperplane-round, and use the limiting aligned-local-field law

$$L = \sqrt{\frac{2}{\pi}}|G_0| + \sqrt{1 - \frac{2}{\pi}}G_1.$$

Independently flipping a negatively aligned site with probability  $\min((-L)_+/2, 1)$  gives the stated gain. The traffic input is of the type developed in [13]; the conclusion above is nonperturbative at fixed algorithmic depth.

There is also a coefficientwise conference free-energy theorem. If  $H = Q_C/\sqrt{n-1}$ , then for every fixed  $r$ ,

$$\frac{\kappa_{2r}(H)}{n} \longrightarrow (-1)^{r-1} \frac{(2r)!}{4r} \text{Cat}_{r-1}.$$

These are the Taylor coefficients of the formal orthogonal-model function

$$G(\beta) = \frac{1}{4} \left( \sqrt{1 + 4\beta^2} - 1 - \log \frac{1 + \sqrt{1 + 4\beta^2}}{2} \right).$$

This is *not* a uniform deterministic free-energy comparison: the Taylor series has radius  $1/2$ , while an improvement over  $\gamma_{\text{loc}}$  would require large  $\beta$ . At degree ten the first class-sensitive invariant appears, but its normalized fixed-order contribution is  $O(1/n)$ ; fixed-order universality survives.

## 5 Structure and arithmetic

**Proposition 5.1** (block control). *For  $[n] = S \sqcup T$ ,*

$$\text{Cut}(A_S) + \text{Cut}(A_T) \leq \text{Cut}(A) \leq \text{Cut}(A_S) + \text{Cut}(A_T) + \|A_{S,T}\|_{\infty \rightarrow 1}.$$

The lower inequality combines two block optimizers and chooses a simultaneous sign flip so that the cross term is nonnegative. The upper inequality decomposes a global optimal signed cut into its internal and cross parts.

Principal restriction and arbitrary one-vertex extension give

$$F(n) \leq F(n+1) \leq F(n) + n.$$

Thus  $|L_{n+1} - L_n| = O(n^{-1/2})$ , and the set of subsequential limits of  $(L_n)$  is a closed interval. In particular,  $H(n) = F(n)^{2/3}$  is nondecreasing. These facts alone do not imply convergence.

## 5.1 Two-block composition and localization

For  $A \in \mathcal{S}_m$ , write

$$f(A) = \max_x |Q_A(x)|, \quad p(A) = \max_x Q_A(x), \quad r(A) = -\min_x Q_A(x),$$

and put  $\Delta(A) = |p(A) - r(A)|$ , so that  $f(A) = \max\{p(A), r(A)\}$ . Given  $B \in \mathcal{S}_n$ ,  $W \in \{\pm 1\}^{m \times n}$ , and  $\sigma \in \{\pm 1\}$ , form

$$P_{\sigma, W} = \begin{pmatrix} A & W \\ W^\top & \sigma B \end{pmatrix}.$$

The two configurations  $(x, y)$  and  $(x, -y)$  give opposite bridge fields. Consequently the following identity is exact:

$$f(P_{\sigma, W}) = \max_{x, y} (|Q_A(x) + \sigma Q_B(y)| + |x^\top W y|). \quad (9)$$

Indeed, this is just  $\max\{|h+b|, |h-b|\} = |h| + |b|$  for each projective pair of states. In the bridge-free internal problem, choosing the relative orientation gives the further exact formula

$$\min_{\sigma=\pm 1} \max_{x, y} |Q_A(x) + \sigma Q_B(y)| = f(A) + f(B) - \min\{\Delta(A), \Delta(B)\}. \quad (10)$$

For  $\sigma = 1$  the left-hand cap before minimization is  $\max\{p(A) + p(B), r(A) + r(B)\}$ ; for  $\sigma = -1$  it is  $\max\{p(A) + r(B), r(A) + p(B)\}$ . Comparing these four sums proves (10), including tied extrema. This saving concerns the internal direct sum only; adding a bridge still requires control of the correlated maximum in (9).

Put

$$R(m, n) = \min_{W \in \{\pm 1\}^{m \times n}} \|W\|_{\infty \rightarrow 1}, \quad \mu_s = \mathbb{E}|\varepsilon_1 + \cdots + \varepsilon_s|,$$

where the  $\varepsilon_i$  are independent uniform signs.

**Theorem 5.2** (rectangular bridge bounds). *For all  $m, n \geq 2$ ,*

$$\begin{aligned} \max\{m\mu_n, n\mu_m\} \leq R(m, n) \leq \min\{ & \sqrt{2(\log 2)mn(m+n-1)}, \\ & m\mu_n + \sqrt{2mn(n-1)\log 2}, \\ & n\mu_m + \sqrt{2mn(m-1)\log 2}\}. \end{aligned}$$

*In particular, if  $n = o(m)$ , then*

$$R(m, n) = m\mu_n(1 + O(\sqrt{n/m})).$$

*Proof.* For fixed  $W$ ,  $\|W\|_{\infty \rightarrow 1} = \max_y \sum_i |(Wy)_i|$ . Averaging over  $y$  gives the lower bound  $m\mu_n$ ; transposition gives  $n\mu_m$ .

For the first upper bound, take  $W$  uniformly at random. For each rank-one sign pattern  $(x_i y_j)_{ij}$ , Hoeffding's inequality bounds the upper tail of  $x^\top W y$ . There are  $2^{m+n-1}$  distinct



patterns, so a union bound gives the first displayed estimate. For the asymmetric estimate, fix  $y$  and put  $Z_y = \sum_i |(Wy)_i|$ . It has mean  $m\mu_n$ , and changing one entry of  $W$  changes  $Z_y$  by at most 2. McDiarmid's inequality, followed by a union bound over the  $2^{n-1}$  projective choices of  $y$ , gives the second upper bound; transposition gives the third. At an equality threshold one applies the argument with an arbitrarily small positive increment and then uses finiteness. Finally, the Khintchine lower bound  $\mu_n \gg \sqrt{n}$  compares the asymmetric error to  $m\mu_n$ .  $\square$

**Corollary 5.3** (unbalanced composition). *Let  $H(n) = F(n)^{2/3}$  and  $N = m+n$ , where  $m \geq n \geq 2$ . Then*

$$H(N) \leq H(m) + H(n) + 1.291\sqrt{nN}. \quad (11)$$

*Thus, for every fixed  $\varepsilon > 0$ , the condition  $n \leq N^{1-\varepsilon}$  gives an  $O(N^{1-\varepsilon/2})$  defect; merely  $n = o(N)$  gives an  $o(N)$  defect.*

*Proof.* Choosing optimal children and a bridge attaining  $R(m, n)$  in (9) gives  $F(N) \leq F(m) + F(n) + R(m, n)$ . Concavity, subadditivity of the  $2/3$  power, and  $F(m) \geq m\sqrt{m-1}/\pi$  from (2) yield

$$H(N) \leq H(m) + H(n) + \frac{2}{3}\pi^{1/3}\sqrt{2\log 2}\left(\frac{m}{m-1}\right)^{1/6}\sqrt{n(N-1)}.$$

The coefficient is at most  $(2/3)\pi^{1/3}\sqrt{2\log 2}2^{1/6} = 1.2904012036\dots < 1.291$ .  $\square$

The all-splits formulation of power-saving composition is stronger than needed. The following localization is the main structural gain.

**Theorem 5.4** (comparable-split localization). *Let  $(b_n)$  be a nonnegative nondecreasing sequence. Suppose there are fixed  $C < \infty$ ,  $\delta > 0$ , and  $n_0$  such that*

$$b_{r+s} \leq b_r + b_s + C(r+s)^{1-\delta}$$

*whenever  $r, s \geq n_0$  and  $1/2 \leq r/s \leq 2$ . Then  $b_n/n$  converges. Consequently, this hypothesis only for comparable splits of  $b_n = H(n) = F(n)^{2/3}$  would prove convergence of  $L_n$ .*

*Proof.* Fix  $k \geq n_0$  and construct a block of size  $qk$  from  $q$  leaves of size  $k$ , recursively splitting each leaf count into its floor and ceiling halves. Every merge is between comparable children. An internal node with  $\ell$  descendant leaves contributes at most  $C(\ell k)^{1-\delta}$ . Charge this equally to those leaves. Along any leaf-to-root path, the first internal size is at least 2 and subsequent sizes grow by a factor at least  $3/2$ . Therefore each leaf receives at most

$$Ck^{1-\delta}K_\delta, \quad K_\delta = \frac{2^{-\delta}}{1 - (2/3)^\delta}.$$

Summing over the leaves gives

$$\frac{b_{qk}}{qk} \leq \frac{b_k}{k} + CK_\delta k^{-\delta}.$$

For arbitrary  $n$ , take  $q = \lceil n/k \rceil$  and use monotonicity  $b_n \leq b_{qk}$ . Letting  $n \rightarrow \infty$  gives the same upper bound for the limsup of  $b_n/n$ . Finally let  $k \rightarrow \infty$  along a subsequence attaining the liminf. The limsup and liminf coincide.  $\square$

## 5.2 Quotient-stratified shell switching

The exact block identity admits a hierarchy between unstratified energy histograms and the full switching response. Let

$$\mathcal{X}_r = \{\pm 1\}^r / \{\pm 1\}, \quad G = \mathcal{X}_m \times \mathcal{X}_n, \quad |G| = 2^{m+n-2},$$

written additively, although its operation is coordinatewise multiplication. Fix  $A, B, \sigma, W$  as above and let  $L = \|W\|_{\infty \rightarrow 1}$ . For  $t = (x, y) \in G$ , define

$$\eta(t) = f(A) + f(B) - |Q_A(x) + \sigma Q_B(y)|, \quad \zeta(t) = L - |x^\top W y|.$$

If  $s = (p, q)$  switches the rows and columns of  $W$ , then

$$f(P_{\sigma, W^s}) = f(A) + f(B) + L - \Gamma(s), \quad \Gamma(s) = \min_{t \in G} \{\eta(t) + \zeta(s+t)\}. \quad (12)$$

Let  $\pi : G \rightarrow Q$  be a quotient homomorphism and put  $K = \ker \pi$ . Retain only the shell counts stratified by quotient class:

$$N_\eta(e, a) = \#\{t : \eta(t) = e, \pi(t) = a\}, \quad N_\zeta(z, a) = \#\{t : \zeta(t) = z, \pi(t) = a\}.$$

**Theorem 5.5** (quotient-shell switching). *If an integer  $g \geq 0$  and a class  $c \in Q$  satisfy*

$$\sum_{e+z < g} \sum_{a \in Q} N_\eta(e, a) N_\zeta(z, c+a) < |K|, \quad (13)$$

*then some switching  $s$  with  $\pi(s) = c$  has  $\Gamma(s) \geq g$ . Hence*

$$f(P_{\sigma, W^s}) \leq f(A) + f(B) + L - g.$$

*The trivial quotient is the ordinary histogram first-moment criterion; the identity quotient is exhaustive switching.*

*Proof.* For a switching  $s$ , let

$$V_g(s) = \#\{t \in G : \eta(t) + \zeta(s+t) < g\}.$$

Average  $V_g(s)$  over the coset  $\pi(s) = c$ . For fixed  $t$  in class  $a$ , the translate  $s+t$  is uniform on the class  $c+a$ . The average is therefore the left side of (13) divided by  $|K|$ . If it is less than one, some integer-valued  $V_g(s)$  is zero, which is equivalent to  $\Gamma(s) \geq g$ .  $\square$

For a one-bit quotient with nontrivial character  $\chi$ , put

$$T_g = \sum_{s \in G} V_g(s), \quad \widehat{V}_g(\chi) = \sum_{s \in G} \chi(s) V_g(s).$$

The two coset sums are  $(T_g \pm \widehat{V}_g(\chi))/2$ . Thus the readily checked condition

$$T_g - |\widehat{V}_g(\chi)| < |G| \quad (14)$$

certifies gain  $g$ . This is the first genuinely state-sensitive level of the hierarchy: the unstratified test retains only  $T_g$ . It does not require the full violation table. If  $f_e = \mathbf{1}_{\{\eta=e\}}$  and  $h_z = \mathbf{1}_{\{\zeta=z\}}$ , the unnormalized Fourier transform gives the exact factorization

$$\widehat{V}_g(\chi) = \sum_{e+z < g} \widehat{f}_e(\chi) \widehat{h}_z(\chi).$$

Thus one Fourier coefficient from each shell suffices for (14).

There is a still coarser two-replica sufficient condition. Define the bad-pair set and its addition map by

$$\mathcal{B}_g = \{(t, u) \in G^2 : \eta(t) + \zeta(u) < g\}, \quad \varphi(t, u) = t + u.$$

Then  $V_g(s) = \#\varphi^{-1}(s)$  and  $T_g = |\mathcal{B}_g|$ . Put  $U_g = \sum_s V_g(s)^2$  and  $S = |G|$ . Parseval gives

$$\sum_{\chi \neq 1} |\widehat{V}_g(\chi)|^2 = SU_g - T_g^2.$$

Consequently, when  $T_g \geq S$ , the inequality

$$SU_g - T_g^2 > (S - 1)(T_g - S)^2 \quad (15)$$

forces a nontrivial character satisfying (14).

The statistic has an exact overlap factorization. For  $r \in G$ , let

$$A_r(e, e') = \#\{t : \eta(t) = e, \eta(t+r) = e'\}, \quad C_r(z, z') = \#\{u : \zeta(u) = z, \zeta(u+r) = z'\}.$$

**Proposition 5.6** (two-replica overlap factorization). *One has*

$$U_g = \sum_{r \in G} \sum_{\substack{e+z < g \\ e'+z' < g}} A_r(e, e') C_r(z, z'). \quad (16)$$

The  $r = 0$  contribution is  $T_g$ ; the other terms count ordered collisions of distinct bad pairs under  $\varphi$ .

*Proof.* Expand  $U_g$  as the number of ordered pairs  $((t, u), (t', u')) \in \mathcal{B}_g^2$  with  $t + u = t' + u'$ . In the elementary two-group  $G$ , this equality is equivalent to  $t + t' = u + u' = r$ . Fixing  $r$  separates the choices of  $t$  and  $u$ , giving (16).  $\square$

There is also a useful integer normal form. Let  $P_g$  be the number of unordered pairs of distinct elements of  $\mathcal{B}_g$  in a common fiber of  $\varphi$ .

**Corollary 5.7** (collision normal form). *Suppose  $T_g = S + q$  with  $q \geq 0$ . Then*

$$U_g = T_g + 2P_g, \quad P_g = \sum_s \binom{V_g(s)}{2},$$

and (15) is equivalent to either of the exact conditions

$$P_g > \binom{q+1}{2}, \quad \sum_s (V_g(s) - 1)^2 > q^2. \quad (17)$$

*Proof.* Each fiber contributes its diagonal pairs and twice its unordered distinct pairs, so  $U_g = T_g + 2P_g$ . Substitution of  $T_g = S + q$  into (15) cancels  $S$  and gives  $2P_g > q(q+1)$ . The centered form follows from  $\sum_s (V_g(s) - 1)^2 = 2P_g - q$ .  $\square$

Thus the global two-replica test asks whether collisions beat the precise cost of the  $q$  bad pairs beyond one per switching. Sharp bounds are already available from the occupancy profile. Put  $Z_g = \#\{s : V_g(s) = 0\}$ ,  $K = S - Z_g$ , and  $M = \max_s V_g(s)$ . Write

$$T_g = \alpha K + \rho, \quad 0 \leq \rho < K, \quad E := T_g - K.$$

Define

$$P_{\min} = K \binom{\alpha}{2} + \rho \alpha.$$

If  $M = 1$ , put  $P_{\max} = 0$ ; otherwise write  $E = \beta(M - 1) + \tau$ ,  $0 \leq \tau < M - 1$ , and put

$$P_{\max} = \beta \binom{M}{2} + \binom{\tau + 1}{2}.$$

**Proposition 5.8** (sharp occupancy bounds). *For every nonnegative integer fiber profile with the stated  $T_g, Z_g, M$ ,*

$$P_{\min} \leq P_g \leq P_{\max}.$$

*The lower bound is best possible from  $T_g, Z_g$  alone, and the upper bound is best possible after also imposing the cap  $M$ . Consequently,  $P_{\min} > \binom{q+1}{2}$  certifies (15), whereas  $P_{\max} \leq \binom{q+1}{2}$  rules it out.*

*Proof.* Among the  $K$  positive fibers with fixed sum, convexity of  $v \mapsto \binom{v}{2}$  makes the collision count smallest when the sizes differ by at most one, giving  $P_{\min}$ . After placing one bad pair in each positive fiber, the remaining  $E$  entries lie in boxes of capacity  $M - 1$ . Convexity makes the collision count largest by filling boxes to capacity and then one final box, giving  $P_{\max}$ . The balanced and concentrated profiles attain the respective bounds.  $\square$

Symmetry can compress the exact overlap sum. If  $\mathcal{G} \leq \text{Aut}(G)$  preserves both  $\eta$  and  $\zeta$ , then  $A_r$  and  $C_r$  are constant on every  $\mathcal{G}$ -orbit of differences. Thus (16) needs one joint overlap signature per orbit, rather than one per element of  $G$ . An asymptotic family whose common stabilizer has only  $2^{o(m+n)}$  difference orbits would therefore give a genuinely compressed exact evaluation of  $U_g$ . The issue is then the inequality, not reconstruction of the full switching table.

The hierarchy also has a sharp multiplicity obstruction.

**Proposition 5.9** (isolated-gain obstruction). *Let  $Z_g = \#\{s : V_g(s) = 0\}$  and suppose every positive  $V_g(s)$  is at least  $v \geq 2$ . If a quotient of dimension  $k$  certifies gain  $g$ , then, with  $d = \log_2 |G| = m + n - 2$ ,*

$$2^{d-k} < \frac{vZ_g}{v-1}. \quad (18)$$

*In particular, if  $Z_g = 2^{o(d)}$ , every quotient-shell certificate for this gain has dimension  $d - o(d)$ .*

*Proof.* A certifying quotient coset has size  $h = 2^{d-k}$  and total violation count less than  $h$ . If it contains  $z \leq Z_g$  zeroes, its other entries contribute at least  $v(h - z)$ . Hence  $v(h - z) < h$ , so  $(v - 1)h < vz \leq vZ_g$ .  $\square$

This identifies a concrete continuation target. A quotient of dimension  $k = o(m + n)$  has only  $2^{o(m+n)}$  stratified classes, so it remains strictly smaller than the full response table. For exact minimizers  $A, B$ , set

$$T = (H(m) + H(n) + C(m + n)^{1-\delta})^{3/2}, \quad g = \max \{0, \lceil F(m) + F(n) + L - T \rceil\}.$$

If an explicit sublinear-dimensional quotient satisfies (13) for this  $g$  on every sufficiently large comparable split, then (12) gives the hypothesis of Theorem 5.4, and  $L_n$  converges. Proposition 5.9 says that this route should target a recurrence-scale gain supported by many switchings, rather than isolated exact parent optima.

The public finite audit uses four fixed optimal-child bridge orbits. The columns give the gain certified by the unstratified histogram, by the best single character, and by exhaustive switching. The final three columns refer to the exhaustive target: its number  $Z$  of successful switchings, the minimum positive violation count  $v$ , and the resulting lower bound on quotient dimension.

| split | $d$ | histogram | one character | exact | $Z$ | $v$ | $k_{\min}$ |
|-------|-----|-----------|---------------|-------|-----|-----|------------|
| 3 + 3 | 4   | 2         | 4             | 6     | 1   | 4   | 4          |
| 3 + 4 | 5   | 4         | 6             | 6     | 8   | 1   | 0          |
| 3 + 5 | 6   | 2         | 4             | 6     | 1   | 2   | 6          |
| 4 + 4 | 6   | 4         | 6             | 8     | 1   | 2   | 6          |

Thus one character improves the certified gain in every displayed case and recovers the exact fixed-orbit cap at  $3 + 4$ . At the other three exact targets, the counting obstruction forces the full quotient dimension. This is finite evidence about the hierarchy, not an asymptotic shell theorem.

The collision audit below is at the displayed one-character target. The column  $R$  is the number of distinct joint overlap signatures in (16);  $q = T_g - S$ ,  $Z = Z_g$ ,  $M$  is the largest fiber, and  $P = P_g$ . The last numeric column is the strict threshold in (17).

| split   | $g$ | $R$ | $q$ | $Z$ | $M$ | $P$ | $\binom{q+1}{2}$ | occupancy |
|---------|-----|-----|-----|-----|-----|-----|------------------|-----------|
| $3 + 3$ | 4   | 4   | 2   | 4   | 2   | 6   | 3                | certifies |
| $3 + 4$ | 6   | 14  | 17  | 8   | 4   | 35  | 153              | rules out |
| $3 + 5$ | 4   | 30  | 11  | 18  | 4   | 36  | 66               | rules out |
| $4 + 4$ | 6   | 15  | 4   | 22  | 2   | 26  | 10               | certifies |

At  $3 + 3$  and  $4 + 4$ , the lower occupancy bound already forces the global two-replica criterion. At  $3 + 4$  and  $3 + 5$ , the sharp bounded-fiber upper bound rules it out, even though a single character succeeds. Thus the unfiltered statistic  $U_g$  can lose a useful coefficient by averaging its energy over all  $S - 1$  nontrivial characters.

This diagnosis suggests a filtered replacement. For any specified family  $\mathcal{C}$  of nontrivial characters,

$$\sum_{\chi \in \mathcal{C}} |\hat{V}_g(\chi)|^2 > |\mathcal{C}|(T_g - S)^2 \quad (19)$$

forces some  $\chi \in \mathcal{C}$  to satisfy (14). It is still a two-replica quantity: with  $K_{\mathcal{C}}(r) = \sum_{\chi \in \mathcal{C}} \chi(r)$ , its left side equals

$$\sum_{s, s' \in G} V_g(s) V_g(s') K_{\mathcal{C}}(s + s').$$

The viable asymptotic target is therefore either a structured lower bound for the collision sum (16), or a  $2^{o(m+n)}$ -sized explicit character family for which the filtered energy does not dilute. The four exact cases distinguish these targets; they do not establish either asymptotically.

The compact finite audit emphasizes why a positive defect is unavoidable. Literal subadditivity of  $H$  fails already at  $2 + 2$ :  $H(4) - 2H(2) = 4^{2/3} - 2 = 0.519842\dots$ . With both children of order at least 3, it first fails at  $3 + 5$  by

$$10^{2/3} - 3^{2/3} - 4^{2/3} = 0.0416629\dots > 0.$$

Among the 42 pairs with  $2 \leq m \leq n$  and  $m + n \leq 15$ , using the evidence-labelled exact-value table, 21 have positive defect, including 15 of the 24 comparable pairs. The largest is  $H(11) - H(5) - H(6) = 1.167629\dots$ . Exhaustive bridge enumeration for four fixed pairs of optimal children also finds orientation sensitivity at  $3 + 3$ : the two relative orientations have exact optimized caps 7 and 5. These are finite checks, not evidence for an asymptotic exponent; the keys, integer sign certificates, and all  $2^{mn-1}$  bridge searches are reproduced by `experiments/composition_small_orders.py`.

The quadratic form satisfies

$$x^T A x \equiv n(n-1) \pmod{4}, \quad F(n) \equiv \binom{n}{2} \pmod{2}.$$

For odd  $n$ , every rectangular cut sum contains an even number of terms, so  $\text{Cut}(A)$  is even.

**Theorem 5.10** (forced unbalancedness). *If  $n \equiv 3 \pmod{4}$ , then*

$$F(n) > \min_{A \in S_n} \text{Cut}(A),$$

*and no  $M$ -minimizing matrix is balanced, where balanced means  $M_+(A) = M_-(A)$ .*

This explains all computed failures of  $F(n) = \min_A \text{Cut}(A)$  at  $n = 3, 7, 11$  and forces another failure at  $n = 15$ , independently of the enumeration.

## 6 Exact computation and provenance

Switching allows the first row and column to be normalized to +1 off the diagonal. The public iterator then enumerates

$$2^{(n-1)(n-2)/2}$$

labeled switching-normalized matrices. For each matrix, global-sign symmetry reduces the Boolean cube to  $2^{n-1}$  vectors. This direct method reproduces orders through 6 and is intentionally simple.

The historical computations through order 15 used a more efficient exact integer pipeline: Gray-code energy updates, principal-submatrix pruning from Lemma 3.1, globally canonical switching/permutation certificates, chunked extension levels, and threshold emptiness combined with the congruence above. The order-15 value  $M = 54$  was certified by two independent empty searches at thresholds 46 and 50, overlapping certificate sets, and direct evaluation of the witness  $[C_{14}, \mathbf{1}]$  on the full antipodal half-cube.

Two implementation failures were discovered and bounded during that work. An early 64-bit certificate encoding overflowed once  $\binom{k-1}{2} > 64$ ; affected levels were recomputed using 128-bit certificates. A fixed-size resume buffer could overflow for sufficiently large seed files; every certified run used smaller per-process seeds, and a patched resume reproduced the relevant levels. These incidents are recorded because they materially affect how the exhaustive claims should be assessed.

The public repository does not ship the multi-gigabyte intermediate levels. Accordingly, entries without a public external reproduction retain a historical-audit status in `results/reported_exact_values.json`; the evidence label on every entry should be read separately.

There is now independent public evidence for part of this range. White’s order-11 package [16], pinned at the revision recorded in `results/external_verification.json`, contains an explicit witness and a standalone C++ lower-bound enumerator. The present audit re-evaluated all  $2^{10}$  projective spin vectors for the witness and reran the lower search: it visited 50,778,686 nodes, completed 936,720 candidates, and found no signing with pair-sum cap 15. This independently proves  $F(11) = 17$ . An external extension witness then gives  $F(12) \leq 18$ ; monotonicity, the order-11 result, and even parity give equality. Six symmetry-complete CP-SAT cases excluding cap 18 at order 13 were also rerun from source, and the order-13 and order-14 witnesses were exhaustively evaluated, corroborating  $F(13) = 20$  and  $F(14) = 21$  [2]. Those CP-SAT runs do not emit standalone proof objects, so their evidence label remains solver-certified rather than formal. None of these external artifacts replaces the historical order-15 threshold audit.

Heuristic search is not a certificate: at order 15, many annealing restarts stalled at  $M = 58$  although the exact optimum is 54.

## 7 The Hermite residual and the remaining absolute-field problem

The squared-Gram proof can be decomposed beyond its first Hermite chaos. Let  $y_\sigma$  be the rounded sign vector for  $X_\sigma$ , write

$$y_\sigma = \sqrt{\frac{2}{\pi}} z_\sigma + \eta_\sigma, \quad R_\sigma = \mathbb{E}[\eta_\sigma \eta_\sigma^\top],$$

and couple the two orientations using one Gaussian vector  $g$ . The arcsine Schur expansion gives, for  $n \geq 5$ ,

$$\frac{R_+ + R_-}{2} \succeq \frac{1 - 2/\pi}{2} I.$$

The paired mean local-field square is at least one, with equality only when  $B^2 = I$ .

Put

$$u = \frac{y_+ + y_-}{2}, \quad v = \frac{y_+ - y_-}{2}.$$

Coordinatewise exactly one of  $u_i, v_i$  is nonzero:

$$(u_i, v_i) = \begin{cases} (\text{sign } g_i, 0), & |g_i| > |(Bg)_i|, \\ (0, \text{sign}(Bg)_i), & |(Bg)_i| > |g_i|. \end{cases}$$

Define the complementary absolute field

$$U_n = \mathbb{E} \sum_i \max\{|(Bu)_i|, |(Bv)_i|\}.$$

The exact remaining target is

$$\liminf_n \frac{U_n}{n} \geq \sqrt{\frac{2}{\pi}}. \quad (20)$$

A self-normalized cube-flip lemma converts (20) into a universal lower-bound improvement to  $0.3233368289\dots$  for  $L_n$ , below the field-plus-spin constant now proved above. Separate second-moment floors  $\mathbb{E}\|Bu\|_2^2, \mathbb{E}\|Bv\|_2^2 \geq n/2$  do not by themselves imply the required first absolute moment.

Although (20) is open universally, it is proved in a large collection of regimes. Write  $D = B^2$  and

$$\bar{\Delta}_n = \frac{1}{n} \sum_{j \neq k} \max\{(n-1)^{-1/2}, |D_{jk}|\}^3.$$

The following statements are classified as **PROVED** within the project's internal audit.

- (i) If  $\bar{\Delta}_n \rightarrow 0$ , then (20) holds. The same remains true after deleting  $o(\sqrt{n})$  exceptional source indices when the complementary cubic mass vanishes.
- (ii) Fixed  $k$ -fold twin blow-ups of a globally decorrelated base satisfy the stronger coefficient  $\sqrt{2/\pi} \sqrt{(k+2)/3}$ . Growing twin fibres are farther from extremality by a direct Boolean lift.
- (iii) Disjoint strong matched pairs, under block-level global decorrelation, satisfy (20).
- (iv) Correlation components of maximum size 265 satisfy (20) under block decorrelation.
- (v) A component partition is unnecessary: a strong-correlation graph of maximum degree 264, with weak off-graph cubic mass, suffices.
- (vi) More generally, with

$$\mathcal{L}(\rho) = \frac{2}{\pi} \left( \arcsin \frac{\rho}{2} - \frac{\rho}{2} \right),$$

the maximum-degree condition may be replaced by the oriented-edge density bound

$$\frac{1}{n} \sum_{(j,k) \in E^{\text{or}}} \frac{|\mathcal{L}(D_{jk})|}{|D_{jk}|} \leq 3.9698962518\dots + o(1).$$

The number 265 is a method threshold, not a conjectured phase transition. The unresolved regime must retain total cubic defect mass  $\Omega(n)$  and a dense or heavily overlapping strong-correlation graph after every admissible weak-cubic decomposition. Simple concentrated candidates such as twins, matchings, bounded components, and bounded-degree giant components have all been eliminated as obstructions.

## 8 Barriers and corrected false routes

The following negative conclusions are part of the research output. They prevent a successor from spending effort on arguments whose quantitative ceiling is already understood.

### 8.1 Spectral and degree-two certificates

Every  $A \in \mathcal{S}_n$  has  $\|A\|_2 \geq \sqrt{n-1}$ . Every diagonal  $Y$  satisfying  $-Y \preceq A \preceq Y$  has  $\text{tr } Y \geq n\sqrt{n-1}$ . For the pair of one-sided semidefinite relaxations,

$$\text{SDP}(A) \text{SDP}(-A) \geq n^2(n-1),$$

so  $\max\{\text{SDP}(A), \text{SDP}(-A)\} \geq n\sqrt{n-1}$ , with equality in the maximum floor exactly for conference matrices. Thus spectral, diagonal-scaling, and degree-two SOS certificates cannot certify the desired constant below the spectral ceiling. The symmetric Grothendieck framework gives the relevant norm dictionary [8].

The tempting one-sided claim  $\text{SDP}(A) \geq n\sqrt{n-1}$  is false. For the order-5 matrix with one positive off-diagonal edge and all other edges negative, exact primal and dual certificates give

$$\text{SDP}(A) = 9,$$

not the earlier numerical estimate 9.008.

### 8.2 Composition, nesting, and regularity

Separate PSD-Gram control of bilinear or Kronecker cross terms loses too much to beat the spectral ceiling. The admissible doubling construction also has ratios that increase under iteration. A generic expression  $A \otimes B + I \otimes S$  need not even be Seidel: some off-diagonal entries are zero.

For a one-vertex extension  $[A, a]$ ,

$$M([A, a]) = \max_y (|y^\top A y| + 2|a \cdot y|).$$

If the maximizers of  $A$  are isotropic, every extension costs at least  $2\sqrt{n}$  asymptotically, forcing a nested chain to coefficient at least  $4/3$ . This applies to several prominent conference-derived seeds.

Switching-regularity alone does not imply spectral attainment. It gives a  $\{\pm 1\}$  eigenvector, but the corresponding eigenvalue need not be extremal. The order-9 rook-graph Seidel matrix is regular with exact  $M = 24 < 27 = 9\|A\|_2$ .

### 8.3 Single-split and superadditive methods

For a rectangular sign matrix, the fractional relaxation of  $\|B\|_{\infty \rightarrow 1}$  equals  $q\mathbb{E}|S_p|$  exactly. Bowlin's theorem is the classical maximum-frustration form of this statement, including its equality and uniqueness structure [3]. Therefore improvement beyond the first moment is a genuine integrality-gap problem, not an averaging or linear duality problem. The rigorous universal ceiling of the single-split route is  $2/(3\sqrt{3}) = 0.384900\dots$ ; the often quoted 0.335 is only numerical.

Cut superadditivity and the associated free energy do not imply convergence. An explicit analytic scalar sequence satisfies the relevant monotonicity, increment, window, and superadditivity properties while oscillating. Fekete normalizations fail on exact small values, and bounded-difference concentration has speed  $n$  where overcoming the number of sign matrices requires speed  $n^2$ .



## 8.4 Several splits and Fourier summaries

The several-splits functional

$$V(A) = \mathbb{E}_u \max_S \sum_{j \notin S} \left| \sum_{i \in S} A_{ij} u_i \right|$$

lies between the single-split mean and  $\text{Cut}(A)$  and is strictly better at several computed orders. It has an exponential  $L^2$  packing, but the desired growing moments do not follow from low-degree Fourier mass.

Several explicit counterexamples delimit this route:

- for every fixed even swap distance  $h$ , there are Seidel matrices for which a split-field increment vanishes identically although the field itself is nonconstant;
- dense, subset-robust level-two Fourier mass can coexist with constant modulus, so all  $L^p$  norms are equal;
- positive shifted absolute kernels have a level-four multiplier whose sign changes at the natural shift scale, ruling out a common level-four Gram sign;
- low-degree mass, coefficient spread, and restriction robustness alone cannot force  $\sqrt{p}$  moment growth.

Any successful continuation must use realizability of the actual shifted absolute-linear-form process or average rigidity over a proportional packed family, not only generic Fourier summaries.

## 8.5 Other closed shortcuts

Graphon convergence is blind because every candidate at the relevant scale converges to the zero graphon. Guerra-style interpolation has no disorder to integrate and convexification admits the zero matrix. Conference local-field arithmetic gives exact finite ceilings but no uniform asymptotic gap; the compatibility equation  $C(Cx) = (n-1)x$  remains essential. Fixed-order conference cumulants agree with the orthogonal model, but that coefficientwise fact does not justify a low-temperature comparison.

## 9 Related formulations

For the induced-subset discrepancy  $D(n)$ ,

$$D(n) \leq F(n) \leq 5D(n).$$

Thus the Erdős–Spencer problem has the same order but not the same constant [5].

The coding formulation is exact. Put  $N = \binom{n}{2}$  and define the augmented cut code

$$\mathcal{C}_n = \{(t + z_i + z_j)_{i < j} : t, z_1, \dots, z_n \in \mathbb{F}_2\} \subseteq \mathbb{F}_2^N.$$

If  $\rho(\mathcal{C}_n)$  is its Hamming covering radius, then

$$F(n) = N - 2\rho(\mathcal{C}_n). \tag{21}$$

Indeed, encode a signing by  $a_{ij} = (-1)^{b_{ij}}$  and a spin vector by  $x_i = (-1)^{z_i}$ . Its pair sum is  $N - 2 \text{wt}(b + (z_i + z_j)_{i < j})$ ; adjoining the complement bit  $t$  accounts for the absolute value, and minimizing over  $b$  takes the covering radius. This is the cut-code connection used in the order-11 certificate [16]. In particular  $F(11) = 17$  is equivalent to  $\rho(\mathcal{C}_{11}) = 19$ , contradicting the

covering-radius formula conjectured by Esmaeili and Zaghian [6]. General weightwise nonlinearity is discussed by Gini and Méaux [9], but no exact weight-two asymptotic constant is known.

For  $B \in \{\pm 1\}^{p \times q}$ ,

$$\min_B \|B\|_{\infty \rightarrow 1} = pq - 2F_{\max}(K_{p,q}),$$

where the right side is maximum frustration of a signed complete bipartite graph. Bowlin gives exact formulas for  $p \leq 7$  [3]; the proportional regime remains open. This is closely related to the Gale–Berlekamp switching game [7, 4, 12], but that game is bilinear and its constants do not directly transfer to the symmetric quadratic problem.

## 10 Open problems and continuation map

**Open problem 10.1** (limit). Does  $\lim_n L_n$  exist? If so, what is its value?

**Open problem 10.2** (power-saving composition). Put  $H(n) = F(n)^{2/3}$ . Prove that there are fixed  $C < \infty$ ,  $\delta > 0$ , and  $n_0$  such that

$$H(m+n) \leq H(m) + H(n) + C(m+n)^{1-\delta}$$

whenever  $m, n \geq n_0$  and  $1/2 \leq m/n \leq 2$ . Theorem 5.4 shows that comparable splits alone make  $H(n)/n$  converge, and hence make  $L_n = (H(n)/n)^{3/2}$  converge. Corollary 5.3 already supplies a power saving when the smaller block is polynomially smaller, but it gives only an order- $m+n$  error on comparable blocks. External composition campaigns identify the same target and reduce one finite-temperature version to controlling a least-likely-output path in a noisy rank-one code [2]. The unresolved obstacle is optimizer-sensitive: the internal block energies and the bridge field are evaluated on the same spin assignment. Theorem 5.5 gives a second precise route: find an explicit quotient of dimension  $o(m+n)$  whose stratified extreme shells certify the recurrence-scale gain. The multiplicity obstruction shows why isolated exact-optimum switchings are the wrong target for that route. Proposition 5.6 reduces the global second moment to simultaneous shell-overlap types, while Proposition 5.8 gives sharp scalar tests for when that moment must succeed or cannot succeed. The finite failures of the global moment despite a successful character show that the alternative is a filtered family as in (19), not another unlocalized average over the entire dual group.

**Open problem 10.3** (absolute field). Prove (20) for the remaining dense overlapping correlation regime, or construct a counterexample. A proof would sharpen the squared-Gram mechanism to  $0.3233368289\dots$ , which is now below the field-plus-spin constant  $\gamma_*$ ; the problem remains structurally useful but is no longer the leading universal-bound route.

**Open problem 10.4** (infinite sub-spectral family). Is there a  $\delta > 0$  and an infinite sequence  $A_n \in \mathcal{S}_n$  such that  $M(A_n) \leq (1-\delta)n^{3/2}$ ?

**Open problem 10.5** (rectangular integrality gap). Determine the proportional asymptotic of  $\min_B \|B\|_{\infty \rightarrow 1}/(q\sqrt{p})$ . The first concrete finite case beyond Bowlin’s formulas is  $p = 8$ , in particular the  $8 \times 16$  problem.

**Open problem 10.6** (order-15 cut). Is  $\min_{A \in \mathcal{S}_{15}} \text{Cut}(A)$  equal to 24 or 26? A structural argument is preferable to repeating the infeasible threshold-96 enumeration. A value-24 witness would have all fifteen order-14 deletions highly unbalanced.

**Open problem 10.7** (Bernoulli lower-tail rate). Does

$$\lim_{n \rightarrow \infty} n^{-2} \log_2 \#\{A \in \mathcal{S}_n : M(A) \leq \lambda n^{3/2}\}$$

exist? Gaussian or typical-scale universality cannot by itself determine this Bernoulli support-endpoint problem.

**Open problem 10.8** (growing-order conference comparison). Can the fixed-cumulant conference universality be upgraded to a deterministic comparison uniform in a growing order or inverse temperature? Any such argument must control the class-sensitive irreducible cores rather than discard them termwise.

## 11 Reproduction guide

The compact public computation is intentionally modest:

```
python -m pip install -e .
python -m unittest discover -s tests -v
python experiments/exact_small_n.py --max-order 6 \
    --output results/exact_small_n.json
python experiments/composition_small_orders.py \
    --output results/composition_small_orders.json
```

It validates Seidel inputs, evaluates  $M(A)$  on the antipodal half-cube, enumerates every first-row-normalized labeled switching representative, computes  $\min_A M(A)$  exactly through order 6, audits all scalar composition defects through total order 15, and exhausts the bridges for four selected fixed-child cases. On four explicit bridge witnesses it also computes every switching violation count, the unstratified and best one-character gains, the quotient-dimension counting obstruction, the exact two-replica collision profile and its sharp occupancy bounds, and the number of distinct simultaneous shell-overlap signatures.

The file `results/external_verification.json` records the two pinned external revisions used in the 2026-08-02 audit, SHA-256 hashes of the relevant sources, exact commands, solver versions, and concise reproduced outputs. The external repositories are intentionally not copied into this release.

The compact report source and the exact-result JSON are intended to be stable handoff points. Large future computations should publish a certificate format, source revision, deterministic regeneration command, hashes, and an independent evaluator before publishing a claimed optimum.

## 12 Final status

The limit problem is unresolved. The strongest universal information is the window  $[\gamma_*, 1/2]$ , with  $\gamma_* = 0.336493364432\dots$ , and exact values through order 15. The field-plus-spin theorem supersedes  $1/\pi$  asymptotically, while the squared-Gram result remains the stronger explicit finite statement of its type. Independent public computations now reproduce order 11 and corroborate orders 12 through 14; the order-15 optimum still rests on the historical threshold audit. For convergence, the cleanest live target is now a power-saving composition bound only for comparable blocks; the localization theorem proves that no all-splits estimate is needed. The quotient-shell hierarchy now supplies an exact intermediate state between unstratified histograms and the full bridge response, while its multiplicity bound rules out compression of isolated gains. The two-replica factorization further shows that common symmetry can compress the required overlap data, but the unfiltered second moment can dilute the decisive character; the live refinement is a structured overlap bound or a subexponential filtered character family. No candidate pair of separated infinite subsequences is known.

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